Asset-Liability Risk and the Relationship Between Balance Sheet and Equity Volatility

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Abstract

This paper examines alternative approaches to measuring risk on a financial balance sheet. Using a stylized Asset-Liability Management (ALM) framework, risk is first defined as the volatility of the net return generated by the difference between asset returns and liability costs, measured relative to total assets. The analysis then considers an alternative perspective in which risk is measured relative to equity. It is shown that, although both approaches are based on the same underlying balance-sheet risks, the equity-based measure incorporates a leverage effect that amplifies volatility as equity becomes a smaller share of total assets. Because leverage magnifies volatility, it can also reduce the long-run growth rate of equity relative to what average returns would suggest. A numerical example illustrates how a balance sheet with moderate asset-based risk may exhibit highly volatile equity returns. The results highlight the distinction between evaluating the risk of a balance sheet as a whole and assessing the sensitivity of its capital position to adverse financial shocks.

1 Introduction

From an ALM perspective, the relevant risk measure is often the volatility of the net return relative to total assets. Such a measure may suggest that balance-sheet risk is moderate and well contained. However, this can be misleading when equity represents only a small fraction of the balance sheet. When equity is small relative to total assets, even modest fluctuations in asset and liability values can generate large changes in equity. In this situation, leverage amplifies volatility, causing the long-run growth of equity to fall well below what the average return would suggest. As a result, conclusions drawn from an asset-based risk measure need not carry over to the capital position of the institution. To illustrate this distinction, the analysis below derives the implied volatility of equity returns and compares it with the corresponding balance-sheet risk measure.

2 Risk Measures

Consider a stylized balance sheet in discrete time with no transactions:

$$A_{t+1} = A_t(1 + r_{A,t}), \tag{1}$$

$$L_{t+1} = L_t(1 + r_{L,t}), \tag{2}$$

where A_t denotes assets, L_t interest-bearing liabilities, $r_{A,t}$ the return on assets, and $r_{L,t}$ the funding cost of liabilities. The institution's net position (economic equity) is defined as

$$N_t = A_t - L_t. \tag{3}$$

Define also liabilities as a share of total assets:

$$\alpha_t = \frac{L_t}{A_t}. \tag{4}$$

2.1 Risk measured relative to assets

From an Asset-Liability Management (ALM) perspective, the relevant risk concept is the risk embedded in the consolidated balance sheet. The change in net worth over one period is

$$\Delta N_t = N_{t+1} - N_t = A_t r_{A,t} - L_t r_{L,t}. \tag{5}$$

Dividing by beginning-of-period assets yields

$$\frac{\Delta N_t}{A_t} = r_{A,t} - \alpha_t r_{L,t}. \tag{6}$$

Define the balance-sheet net return as

$$r_{N,t} = r_{A,t} - \alpha_t r_{L,t}. \tag{7}$$

Since both asset returns and liability costs are stochastic, $r_{N,t}$ is also stochastic. Suppressing time subscripts for notational simplicity, the variance of the balance-sheet net return is

$$\mathrm{Var}(r_N) = \sigma_A^2 + \alpha^2 \sigma_L^2 - 2\alpha \rho_{AL} \sigma_A \sigma_L, \tag{8}$$

where σ_A and σ_L denote the volatilities of asset returns and liability costs, respectively, and ρ_{AL} their correlation. When liabilities account for nearly all assets, i.e. $\alpha \approx 1$, this expression simplifies to

$$\sigma_N^2 = \sigma_A^2 + \sigma_L^2 - 2\rho_{AL} \sigma_A \sigma_L. \tag{9}$$

If asset and liability returns are perfectly correlated, balance-sheet risk is reduced through natural hedging. In the special case where $\alpha = 1$ and $\sigma_A = \sigma_L$, the balance sheet is perfectly matched and

$$\sigma_N = 0. \tag{10}$$

More generally, the correlation coefficient ρ_{AL} may vary through time. Assets and liabilities often differ in duration, implying that shifts in the yield curve can alter the covariance structure between asset returns and liability costs.

2.2 Risk measured relative to equity

The preceding analysis evaluates risk relative to total assets. An alternative perspective is to measure risk relative to equity itself. The return on equity is defined as

$$r_{E,t} = \frac{N_{t+1} - N_t}{N_t}. \tag{11}$$

Using the definition of net worth,

$$N_t = A_t - L_t, \tag{12}$$

together with

$$N_{t+1} - N_t = A_t r_{A,t} - L_t r_{L,t}, \tag{13}$$

gives

$$r_{E,t} = \frac{A_t r_{A,t} - L_t r_{L,t}}{A_t - L_t}. \tag{14}$$

Substituting $\alpha_t = L_t / A_t$ yields

$$r_{E,t} = \frac{r_{A,t} - \alpha_t r_{L,t}}{1 - \alpha_t} = \frac{r_{N,t}}{1 - \alpha_t}. \tag{15}$$

Thus, the return on equity equals the balance-sheet net return scaled by the inverse of the equity ratio. Suppressing time subscripts, the variance of equity returns becomes

$$\mathrm{Var}(r_E) = \frac{1}{(1-\alpha)^2}\,\mathrm{Var}(r_N). \tag{16}$$

Equivalently,

$$\sigma_E = \frac{\sigma_N}{1-\alpha}. \tag{17}$$

Substituting the expression for σ_N gives

$$\sigma_E = \frac{\sqrt{\sigma_A^2 + \alpha^2\sigma_L^2 - 2\alpha\rho_{AL}\sigma_A\sigma_L}}{1-\alpha}. \tag{18}$$

The underlying balance-sheet risk is identical in both approaches. However, measuring risk relative to equity introduces a leverage factor equal to

$$\frac{1}{1-\alpha}. \tag{19}$$

As liabilities approach total assets and equity becomes small relative to the balance sheet,

$$\alpha \to 1, \tag{20}$$

implying

$$\sigma_E \to \infty, \tag{21}$$

provided that $\sigma_N > 0$. A level of volatility that appears moderate when measured relative to assets may therefore correspond to substantial volatility when measured relative to equity.

2.3 Numerical illustration

To illustrate the distinction between balance-sheet risk and equity risk, consider a stylized balance sheet with assets $A = 100$ and liabilities $L = 90$. The leverage ratio is therefore

$$\alpha = \frac{L}{A} = 0.9, \tag{22}$$

implying equity of

$$N = A - L = 10. \tag{23}$$

Assume asset-return volatility $\sigma_A = 8\%$, liability-cost volatility $\sigma_L = 2\%$, and correlation $\rho_{AL} = -0.9$. The volatility of the balance-sheet net return is

$$\sigma_N^2 = \sigma_A^2 + \alpha^2\sigma_L^2 - 2\alpha\rho_{AL}\sigma_A\sigma_L \tag{24}$$

$$= 0.009316, \tag{25}$$

so that

$$\sigma_N = \sqrt{0.009316} = 9.65\%. \tag{26}$$

Since the equity ratio equals

$$1 - \alpha = 0.1, \tag{27}$$

the volatility of equity growth becomes

$$\sigma_E = \frac{\sigma_N}{1-\alpha} = \frac{0.0965}{0.1} = 96.5\%. \tag{28}$$

Although balance-sheet risk is below 10% when measured relative to assets, the corresponding equity volatility is almost 100%. The reason is that equity represents only a small fraction of the balance sheet, implying a leverage multiplier of

$$\frac{1}{1-\alpha} = 10. \tag{29}$$

The example illustrates how moderate balance-sheet risk can translate into highly volatile equity outcomes when capital buffers are small.

2.4 Volatility drag and the distribution of equity outcomes

The leverage effect discussed above has implications that extend beyond the measurement of risk. Because leverage amplifies equity volatility, it also affects the rate at which equity compounds over time. Thus, leverage influences not only the volatility of equity returns but also the long-run growth dynamics of equity. Consider the stylized balance sheet introduced above. Even when the expected return on assets exceeds the funding cost of liabilities, fluctuations in asset and liability values may reduce the geometric growth rate of equity. This phenomenon is commonly referred to as *volatility drag*. For a multiplicative process with arithmetic mean return μ and return volatility σ, a second-order approximation to the long-run geometric growth rate is

$$g \approx \mu - \frac{1}{2}\sigma^2, \tag{30}$$

where μ denotes the arithmetic mean return. Since

$$\sigma_E = \frac{\sigma_N}{1-\alpha}, \tag{31}$$

the increase in equity volatility induced by leverage also increases the magnitude of volatility drag.[1] As leverage rises, an increasingly large fraction of expected return is offset by the adverse effects of volatility on long-run compound growth. To illustrate this effect, Monte Carlo simulations were conducted for the stylized balance sheet with assets of 100, liabilities of 90, and equity of 10. Asset returns were assumed to have a mean of 2.25% and a volatility of 8%, while liability costs were assumed to have a mean of 2% and a volatility of 2%. The correlation between asset and liability returns was set to -0.9. The balance sheet was simulated over a five-year horizon using 500,000 independent scenarios.

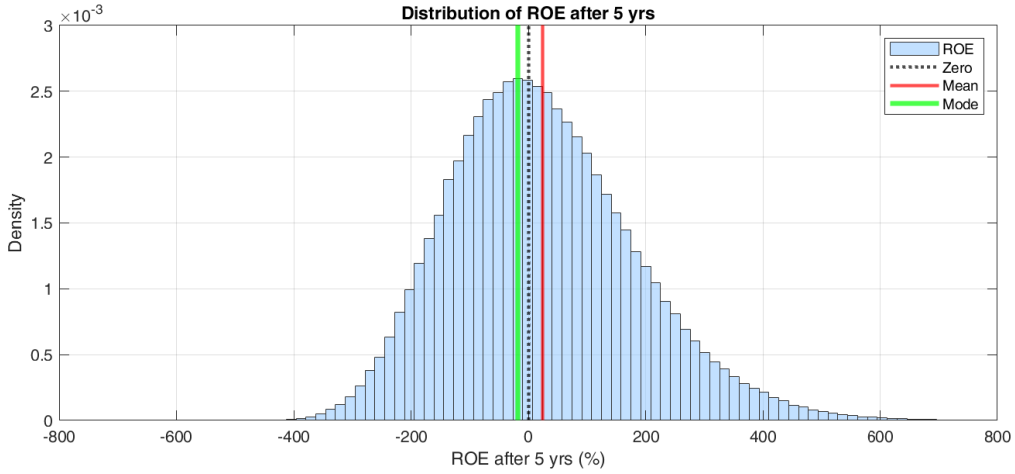


Figure 1: Simulation of ROE

Figure 1 illustrates how leverage increases the positive skewness of the distribution of equity outcomes. A relatively small number of highly favourable realizations generate substantial increases in equity value, whereas a considerably larger share of the probability mass remains concentrated in less favourable regions of the distribution. As leverage rises, the distribution therefore becomes increasingly asymmetric. This asymmetry is reflected in the divergence between alternative measures of central tendency. Although the mean remains positive, owing to the influence of a limited number of exceptionally favourable outcomes, the mode is negative and located in the region of highest probability density. Consequently, the expected outcome

[1] For a lognormal distribution, skewness is $\gamma_1 = (e^{\sigma^2} + 2)\sqrt{e^{\sigma^2} - 1}$. Since γ_1 is a monotonically increasing function of σ, greater volatility is associated with a more positively skewed distribution.

suggests positive equity growth, whereas the most likely outcome corresponds to a decline in equity value.

The increasing asymmetry of the distribution has important implications for the interpretation of summary statistics. In a highly skewed distribution, measures of central tendency may provide limited information regarding the location of the probability mass. The median, for example, partitions the distribution into two equally probable regions but need not coincide with the region associated with the highest probability density. Consequently, neither the mean nor the median necessarily provides an adequate characterization of the economic consequences of volatility drag. A more complete assessment therefore requires consideration of the distribution as a whole.

3 Conclusion

The balance-sheet approach and the equity approach capture different dimensions of financial risk. While the former measures risk relative to total assets, the latter measures its impact on the capital available to absorb losses. Consequently, a level of balance-sheet risk that appears moderate may correspond to substantial volatility relative to equity when leverage is high. Leverage influences not only the volatility of equity returns but also the distribution of equity outcomes, implying that expected outcomes alone may provide an incomplete picture of risk. The appropriate measure therefore depends on the question being asked. A balance-sheet perspective is useful for assessing overall economic risk, whereas an equity-based perspective is more informative when evaluating capital resilience. Together, the two approaches provide complementary insights.